

NFL Sack Probabilities– Summary

- **Context:** This project aims to predict the probability of whether the next pass play will result in a sack or not using play-by-play data from the 2021-2023 NFL Seasons. Using different machine learning models, such as Logistic Regression, Gradient Boosting, Random Forest, Naïve Bayes, and Support Vector Machines, the goal is to assess the data, build a predictive model, and then evaluate the model and quality of its predictions.
- **Reasoning Behind It:** In the NFL, sacks play a huge part in stalling drives. They lead to fewer scoring probabilities and a potential end of offensive possessions. Having a sense of when a sack may occur will help future coaches, quarterbacks, and betting markets optimize their planning and adjustments. This is a great classification problem, whether that be in the realm of betting markets, sport analytics, or simply to further develop team strategies.
- **Machine Learning Model:**
 - *Feature Engineering:* Creating cumulative and rolling (last 8 weeks) sack rate metrics for each respective defense and quarterback was pivotal for identifying long and short-term trends in this model. This context gave important background to each play before being processed in the model in consideration to both team and player tendencies.
 - *Different Models used:*
 - **Logistic Regression:** This model predicts the likelihood of a sack as a function of a linear combination of the input variables.
 - **Gradient Boosting:** The Gradient Boosting model uses decision trees (built in a series), where each subsequent trees corrects the errors of the previous one. The collection of weak models are built into a single accurate prediction model.
 - **Random Forest:** A Random Forest Model uses the combination of multiple decision trees to make a well-rounded prediction.
 - **Naïve Bayes:** A probabilistic classifier model based on Bayes Theorem, where all features are assumed as independent.
 - **Support Vector Machines:** The SVM model will attempt to find the differentiator between sacks and non-sacks using maximum class margin.
 - *Feature Importance:* A feature important output was created using the Logistic Regression model to identify the features that influence sack likelihood/probability the most. The feature importance output showed that 'career_sack_rate_qb_prev' (QB Career sack rate up to that play) had the largest impact.
- **Model Evaluations:**
 - The logistic regression model performed the best in terms of log loss and brier score. It is important to keep in mind that although AUC is generally used to evaluate binary classification problems, it was not as useful here as the 'sack' variable is zero-inflated (sacks only happen a small percentage of plays). Log loss in this case helps identify and measure the accuracy of predicted probabilities.
- **Future Implications:**
 - How could player tracking data affect and improve the model?
 - How could incorporating weather or environment factors affect and improve the model?